Costing the access network — an overview

P Bell, R Walling and J Peacock

This paper highlights the subject of cost modelling of the access network, with particular emphasis on planning and design. A number of issues are discussed, including activity-based costings, and whole-life costs. The paper indicates the methods that may be applied and illustrates some of the current analysis being pursued to investigate optimised network design strategies for fibre in the access network.

1. Introduction

In order to illustrate the reason and significance of cost modelling it is beneficial to consider the life cycle of the design and implementation of a telecommunications network. Figure 1 represents a schematic of the distinct stages of a telecommunications network design and implementation.

As can be seen in Fig 1, costing analysis is required at every stage of the life cycle. Initially, during the requirements capture phase, data collation and analysis is required not only to aid the decision of technology choice, but also to assist the business case and investment appraisal of the proposed network solution.

Cost studies are also performed during the specification, procurement and approval phase of the life cycle, in order to balance the specification against the cost of the chosen solution. For instance, a network may be specified with a number of criteria indicated during the statement of requirements. However, the extent to which each of these criteria is met may have a profound effect upon the final cost. Furthermore, cost studies will indicate competitive prices and cost sensitivities for the procurement of the network components, and aid the portfolio management.

Once the network solution has been approved the network design and planning rules must be developed. This process involves optimising, broadly, how the network is deployed. Analysis of cost sensitivities and key cost drivers is performed [1, 2]. This represents an integral part of the life cycle; a successful deployment strategy will take account of key cost drivers to minimise the expense to the network provider. Failure to provide an adequate deployment strategy can undermine previous work and induce additional network expense, and result in the original business objectives not being achieved.

The final stage is the in-service phase and requires a tarriffing policy. While the market price of a service may not be exclusively dependent upon the supply cost, an understanding of the costs is fundamental to maintaining a competitive advantage [3].

All these stages are influenced by many factors as indicated in Fig 1, including network strategy, process development and simplification, and future network evolution. However, the main driver for any network activity is the business and marketing requirements. Customer requirements and objectives, together with an investment appraisal and business plan, are the key initiators of any network activity [4—6].

Thus, the study of network costs is critical at every stage of a network's life cycle. The remainder of this paper details the methodologies and techniques that can be used to analyse costs, with particular emphasis upon the planning rule development and network design stage of the life cycle.

2. Aspects of cost

An investigation of access network costs may be split into several areas, with each area considered separately. However, in order to compare different solutions and scenarios the whole network cost must be considered. For example, the costs associated with terrestrial network solutions may be split broadly into two categories — electronics costs (e.g. linecards) and physical plant costs (e.g. cable, ducts and nodes). While these two categories may be studied separately, there are significant inter-relations, and therefore they must be considered together to reveal the true overall cost picture.

Similarly, any complete cost study must consider the whole-life costs of the network, rather than just consider the initial deployment cost. The whole life costs include costs

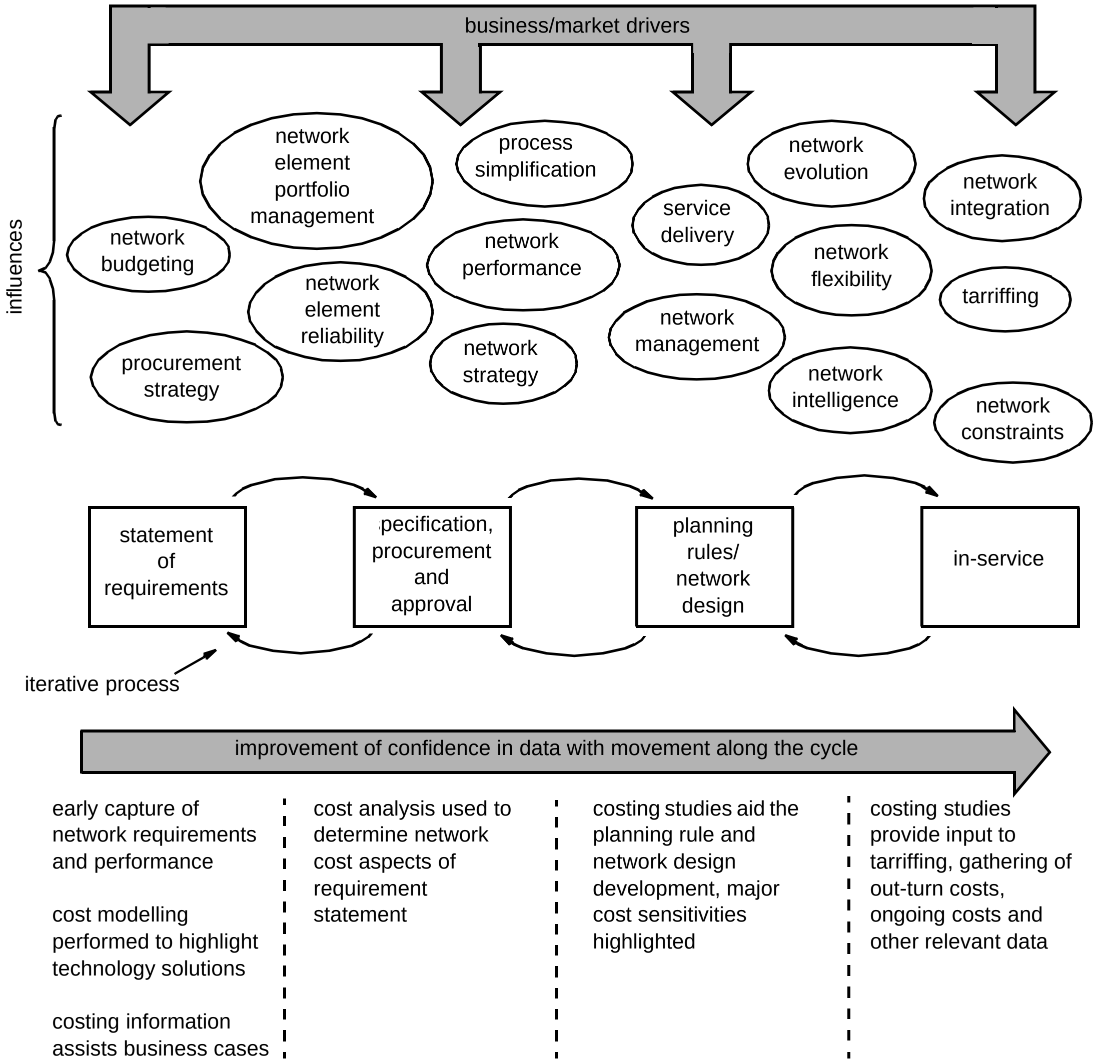

Fig 1 The life cycle of a telecommunications network design and implementation.

associated with each phase of the life cycle (see Fig 1), and take into consideration issues such as the method adopted for the financial accounting (e.g. net present value analysis), network reliability including maintenance and repair, sparing costs, and removal costs at the end of a network's useful life [7, 8].

A thorough analysis should allocate its costs using activity-based costing methodology [9—12]. Activity-based costing (ABC) represents a methodology of accounting for costs based upon capturing all the operations and resources required in order to supply a service. ABC analysis includes taking account of development, deployment, and operating costs to maintain a specific service, i.e. ABC methods would incorporate cost expended during all the phases of the life cycle. In this way, ABC benefits from providing an accurate depiction of the costs in providing a particular service in the access network. However, by the very nature of activity-based costing it is very complex and difficult to collate all the required information. Moreover, difficulty can be encountered in allocating the costs to specific services. For example, consider a fibre cable which provides KiloStream, MegaStream, and PSTN, as well as providing spare capacity for future service requirements. The costs of the cable may be allocated in several ways — by bandwidth or by fibres used. Also, the allocation of costs adds a further complication — how is account taken for that portion of the cable which has been allocated for future service requirements?

Fortunately, depending on which phase of the life cycle is being considered, a number of assumptions may be applied that reduces significantly the problem of costing. Considering specifically the planning rule and network design phase, a number of assumptions may be invoked. The major assumption is that since the development of technology has already occurred, and interest is specifically in the planning rules, then these costs may be neglected. Further, since again the interest is specific to the planning rules for deployment of a particular technology, a number of other costs, e.g. storage of equipment, some transport costs on deployment, some planning time, can be neglected. In this way, the problem may be simplified to a manageable level and tailored to a specific requirement.

These assumptions may apply not only in the development of a particular technology (e.g. fibre), but when considering deployment choices between two similar technologies (e.g. terrestrial technology, copper versus fibre deployment).

3. Network costing methodology

Having considered all the aspects of costing the access network, as described in section 2, the methodology of cost investigation must be chosen. There are two main methods of determining the cost of specific networks [13]. The first method represents the study of an actual proposed network scenario based on a real geographical situation. This involves investigating specific locations and considering the network costs associated with the proposal. This may include surveying duct and route maps for terrestrial network solutions. The main advantage of this method is that the network costs calculated would be accurate, especially the initial deployment costs. However, this costing process can become very time consuming. Moreover, the scenario considered may not be representative of national or global scenarios, and thus the analysis performed is only applicable to the situation considered.

The second method represents the development of mathematical models. Creating a flexible adaptable model allows a simple method of studying a host of different network proposals (see section 4). The main disadvantage of this approach is that it may not be as accurate as the first method for specific network situations. However, a model is a powerful method to investigate the network costs, sensitivities and drivers. Furthermore, with the increased interest in computer aided design (CAD) and computer aided modelling (CAM), a multitude of software tools are available to simplify the modelling process [14].

4. The fibre infrastructure model

A number of models have been developed at BT Laboratories (BTL) to consider all aspects of cost in the access network. One of these models, the fibre infrastructure model, considers the deployment cost of fibre, using OTIAN® technology in the access network [13, 15—17]. This model has been specifically designed to investigate and optimise fibre network design with respect to the physical external plant deployment costs only.

The model is based upon a generic fibre architecture, offering a possible N-level split network as shown in Fig 2. The model may be described as consisting of broadly three sections. Figure 3 shows a schematic of the model with the three main sections, namely the inputs and outputs section, the calculation section, and the elements section.

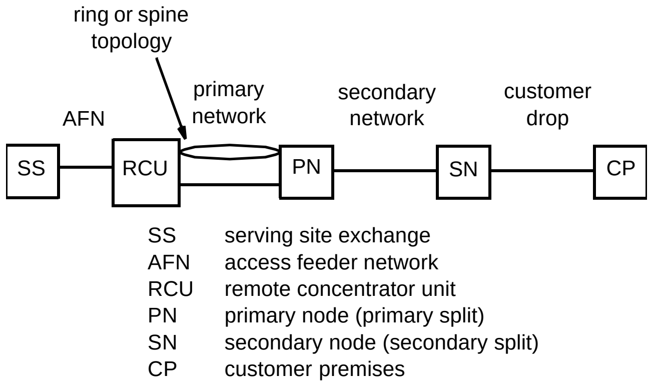

Fig 2 Generic fibre architecture.

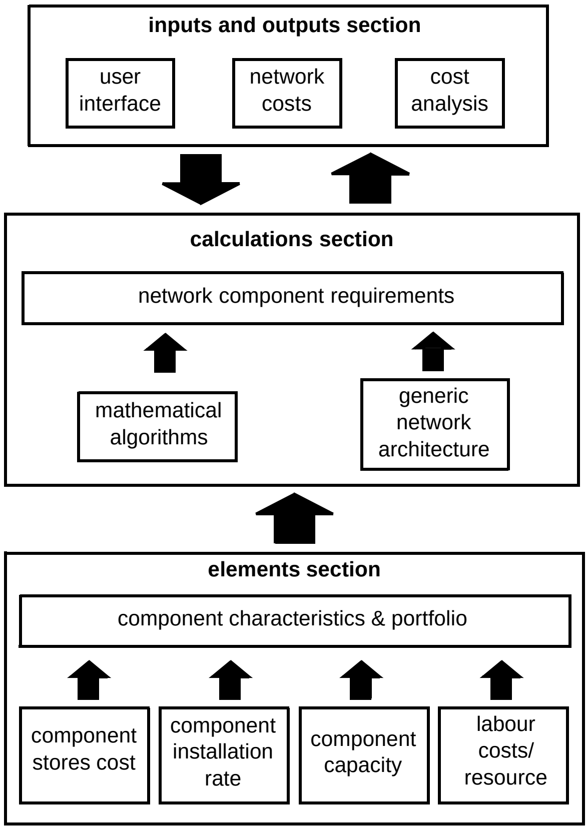

Fig 3 The schematic of the fibre infrastructure model.

As detailed in Fig 3, the inputs and outputs section incorporates a user interface so that the user can input details of the desired scenario to be modelled, while also providing an output of the network deployment cost information including analysis of the key cost components. The elements section provides the model with data regarding the fibre network components. This data includes the price, the installation cost and the capacity of the components, creating a portfolio of component characteristics. The calculation section contains mathematical algorithms which combine the inputs and outputs section with the elements section to determine the quantities of each component required.

Figure 4 illustrates an example of a typical output of the model. This particular output was used to assist the development of planning rules for fibre deployment. The costs associated with deploying 32-way PON architectures in the access network are detailed in Fig 4. The impact of penetration (take-up of service) is considered and comparisons between the different 32-way PON configurations are highlighted. Figure 4 illustrates that the use of a 2 × 4 splitter as the primary split and a 2 × 8 splitter as the secondary split provide an optimum cost solution over the range of 20% to 80% penetration.

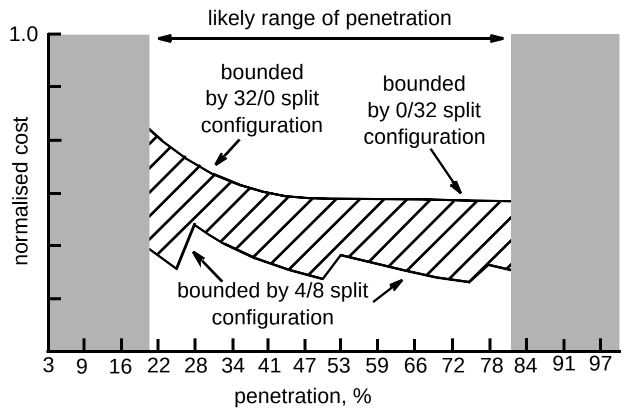

Fig 4 The effect of penetration (take-up) upon fibre infrastructure costs, where n_1/n_2 refers to a first level of split using $1 \times n_1$ splitter, and a second level of split using a $1 \times n_2$ splitter.

Once the split configuration has been studied the other cost sensitivities may be considered. One aspect that may be of interest is the spread of the costs across the different network locations (see Fig 2). This is shown in Fig 5, which illustrates how the costs are distributed over the network. Figure 5 highlights the stores or component costs and the labour costs to install the network. As can be seen, the costs increase as the network gets closer to the customer, because of the reduction in sharing of the infrastructure.

It is analysis of this nature that aids the process of optimising the network costs, by implementing the appropriate planning and design rules. However, as mentioned in section 2, the infrastructure deployment costs must not be considered in isolation. The whole-life and whole network costs must be incorporated in any proposed

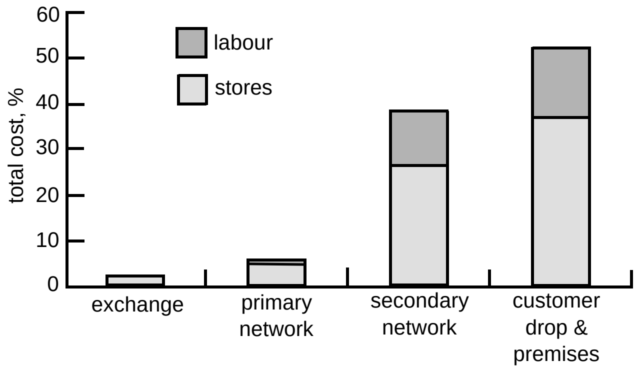

Fig 5 The cost spread across the different network locations.

planning and design rules. To this end, reliability analysis of the proposed network, integrated with details of the associated operations and maintenance processes, will result in an assessment of the operating, maintenance and sparing costs. This information combined with the initial deployment can give an indication of the whole-life cost.

5. Conclusions

This paper has put into context the role of network costing during the life cycle of a telecommunications network design and implementation. Several main issues regarding network cost modelling and their significance for the access network in particular have been discussed, with particular emphasis upon the use of cost modelling when considering network design and planning rule analysis. Two possible methodologies for considering network costs have been highlighted, and the effectiveness of mathematical modelling underlined.

The paper gave an example of mathematical modelling of capital costs, by describing the fibre infrastructure model currently in use at BTL. A specific result of the model was considered, reiterating the importance of cost modelling in the network design phase of the life cycle.

References

1 Lea-Wilson N D: ‘Key factors governing cost-effective deployment of passive optical infrastructures’, Second Communication Networks Symposium, Manchester Metropolitan University, pp 102—105 (July 1995).

2 Dalgoutte D G, and Lea-Wilson N D: ‘Installation cost issues in external plant deployment’, 13th Annual Conference on European Fibre Optic Communications and Networks proceedings 3, (June 1995).

3 Franklin B: ‘Pricing and profitability in telecommunications’, Telecommunications Engineering, A Structured Information Programme, 3.2 (July 1995).

4 Pilgrim L: ‘Investment appraisal’, Telecommunications Engineering, A Structured Information Programme, 4.4 (July 1993).

5 Schraer M: ‘Business planning and modelling’, Telecommunications Engineering, A Structured Information Programme, 4.5 (April 1994).

6 Sharpe M, Guyon R and Lisle P: ‘The evolving structure of the access network’, Telecommunications Engineering, A Structured Information Programme, 10.1 (October 1995).

7 Davies N J and Napier I: ‘New directions in investment appraisal’, BT Technol J, 12, No 4, pp 42—51 (October 1994).

8 Andersen N E, Eriksson M, Gardan D, Margarido R, Redstal M, Socard A, Steffensen H, Stroetzel J: ‘Life cycle cost for PON systems’, 6th IEEE Workshop on Optical Access Networks (October 1994).

9 Bussey B A: ‘ABC within a service organisation’, Management Accounting, Dec. 1993.

10 Gwynne R and Ashworth G: ‘Implementing activity-based management at Mercury Communications’, Management Accounting, (December 1993).

11 Booth R: ‘Life-cycle costing’, Management Accounting (June 1994).

12 Jeffries A and Hankes J: ‘Back to basics?’, Management Accounting (June 1994).

13 Hale P G, Peacock J and Bell P: ‘Modular optical plant for the access network: a practical solution’, 13th Annual Conference on European Fibre Optic Communications and Networks proceedings, 3 (June 1995).

14 Asumu D E and Senior J M: ‘Software package for designing/planning networks in the subscriber loop’, Fibre and Integrated Optics J, 15, No 1 (January 1996).

15 Brewer D A, Hale P G, Peacock J, Walling R: ‘Modular optical plant for access network: operational aspects’, 13th Annual Conference on European Fibre Optic Communications and Networks proceedings, 3 (June 1995).

16 Dalgoutte D, Hewins A, Hornung S and Peacock J: ‘OTIAN — an outside plant system for the access network’, 13th Annual Conference on European Fibre Optic Communications and Networks proceedings, 3 (June 1995).

17 Frost P, Peacock J: ‘OTIAN — an infrastructure system for networks in the future’, Communications Symposium, Manchester Metropolitan University, pp 97—100 (July 1994).

Peter Bell graduated from Coventry Polytechnic in 1990 obtaining first class honours degree in Mathematics. In 1993, he graduated from Newcastle University with a PhD in Mathematics.

Since joining BT Laboratories in 1993, he has worked in the external plants and pilots unit, within the network transport engineering centre. This work has involved analysing the performance of fibre and copper in the access network. In particular, the work has aided the development of fibre deployment planning rules.

He is a member of the Institute of Mathematics and its A pplications, and the Institute of Electrical Engineers

Richard Walling has worked for BT since 1972 and during this time has been awarded a BA (Hons) and a certificate in management, both with the Open University.

For 20 years he was employed on materials research tasks particularly epitaxial growth and analysis of semiconductor material for optoelectronic devices. He is now working on system and component reliability and cost analysis of the access network infrastructure.

He is an Incorporated Engineer and a Member of the Institute of Engineers and Technicians.

John Peacock joined BT in 1980 in Leeds after spending 5 years in the communications branch of the Royal Navy. In 1983 he attended Leeds Polytechnic and graduated in 1987 with a first class honours in Communications Engineering. He then joined BT Lab-oratories and has worked on a number of access network related projects, including leading the teams developing the TPON specifications, OTIAN® plant system, access network cost and performance models, and access network planning rules. He currently leads a team which specialises in access network infrastructure performance and planning issues, including the development of advanced planning tools. He holds an MBA and is a Chartered Engineer and a Member of the IEE